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(54) **INK LIQUID LEVEL HEIGHT MEASURING APPARATUS AND IMAGE FORMING APPARATUS**

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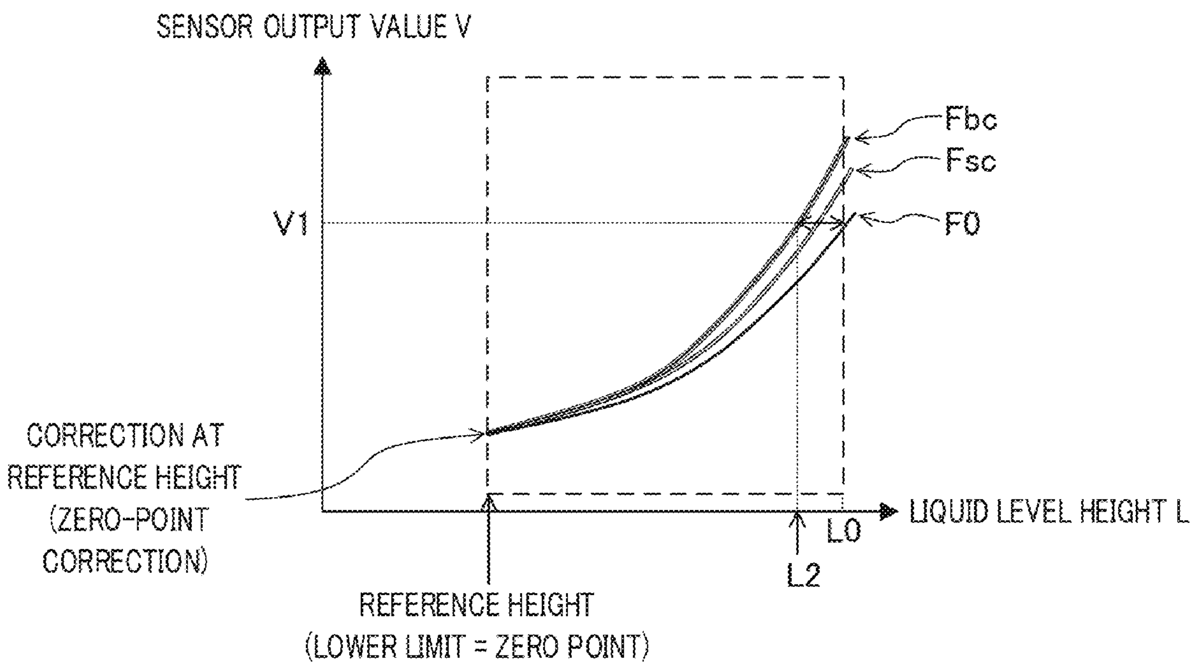
(57) **ABSTRACT**

Provided is an ink liquid level height measuring apparatus that measures a liquid level height of ink in a container. The ink liquid level height measuring apparatus includes: a float that includes a magnet and moves in association with a change in the liquid level height; a Hall element that detects, as a continuous value, a change in magnetic flux density due to the magnet that moves together with the float; and at least one hardware processor. The at least one hardware processor adjusts the liquid level height to a reference height, acquires an output value of the Hall element, when the liquid level height is the reference height, and corrects the liquid level height based on the output value.



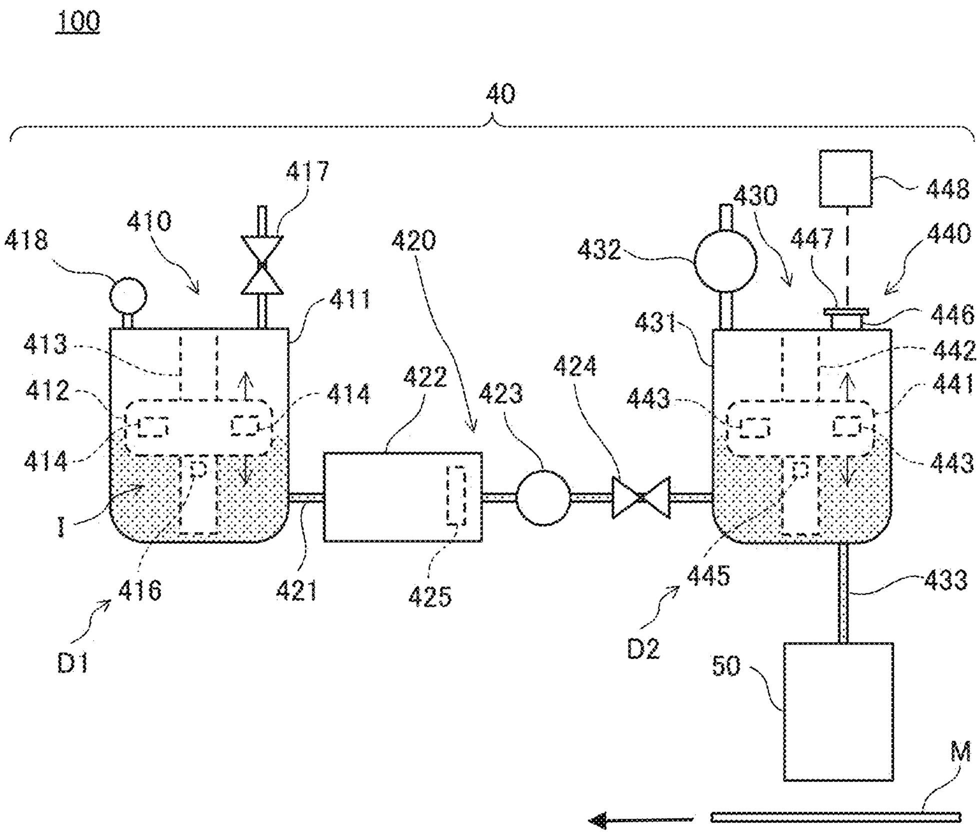

100
40
417
410
418
411
413
412
414
414
I
416
D1
420
422
423
424
421
425
430
432
431
448
447
440
446
442
441
443
443
445
D2
433
50
M

FIG. 1

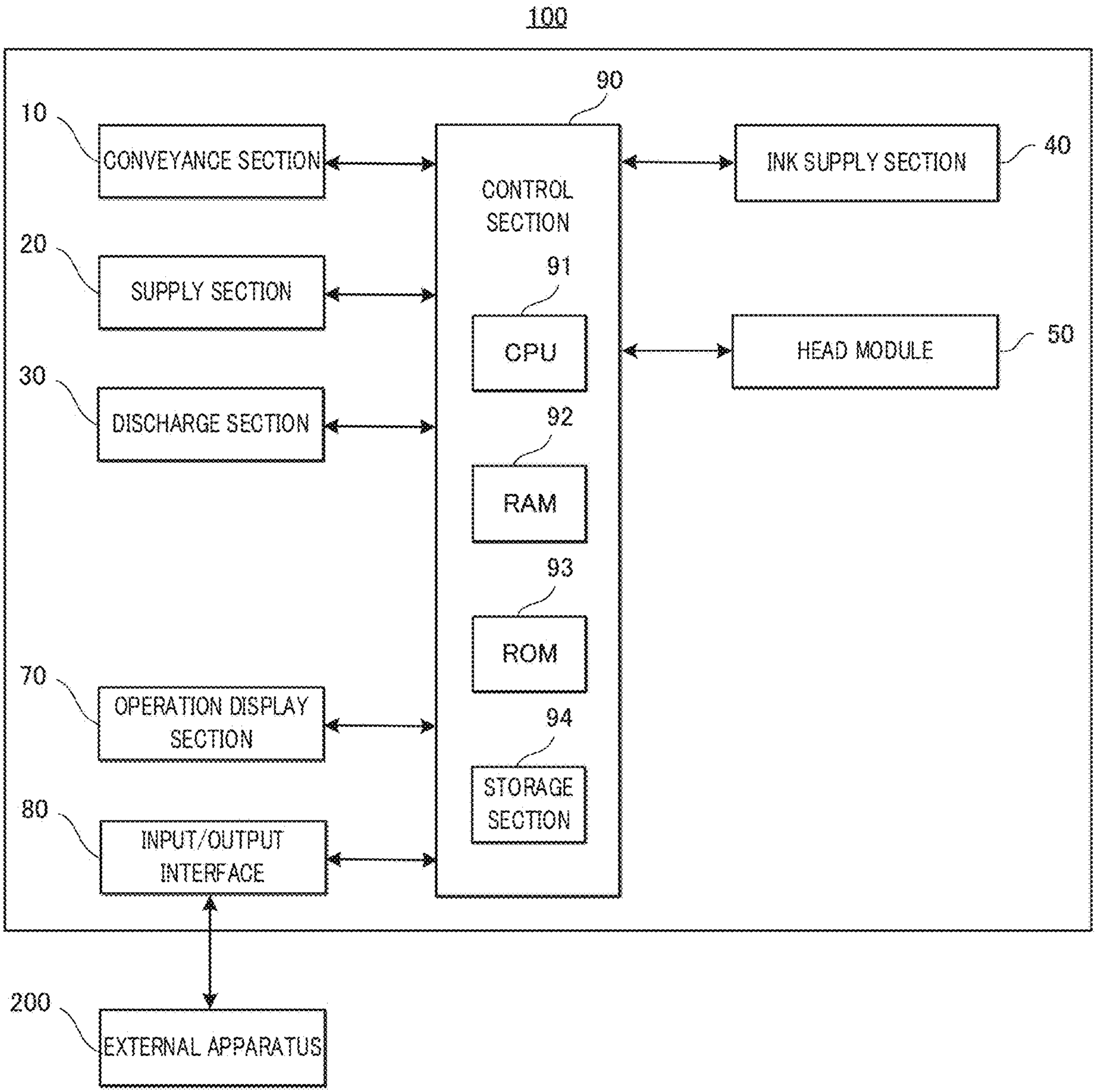

100
90
10
CONVEYANCE SECTION
CONTROL
SECTION
INK SUPPLY SECTION
40
20
SUPPLY SECTION
91
CPU
HEAD MODULE
50
30
DISCHARGE SECTION
92
RAM
93
ROM
70
OPERATION DISPLAY
SECTION
94
STORAGE
SECTION
80
INPUT/OUTPUT
INTERFACE
200
EXTERNAL APPARATUS

FIG. 2

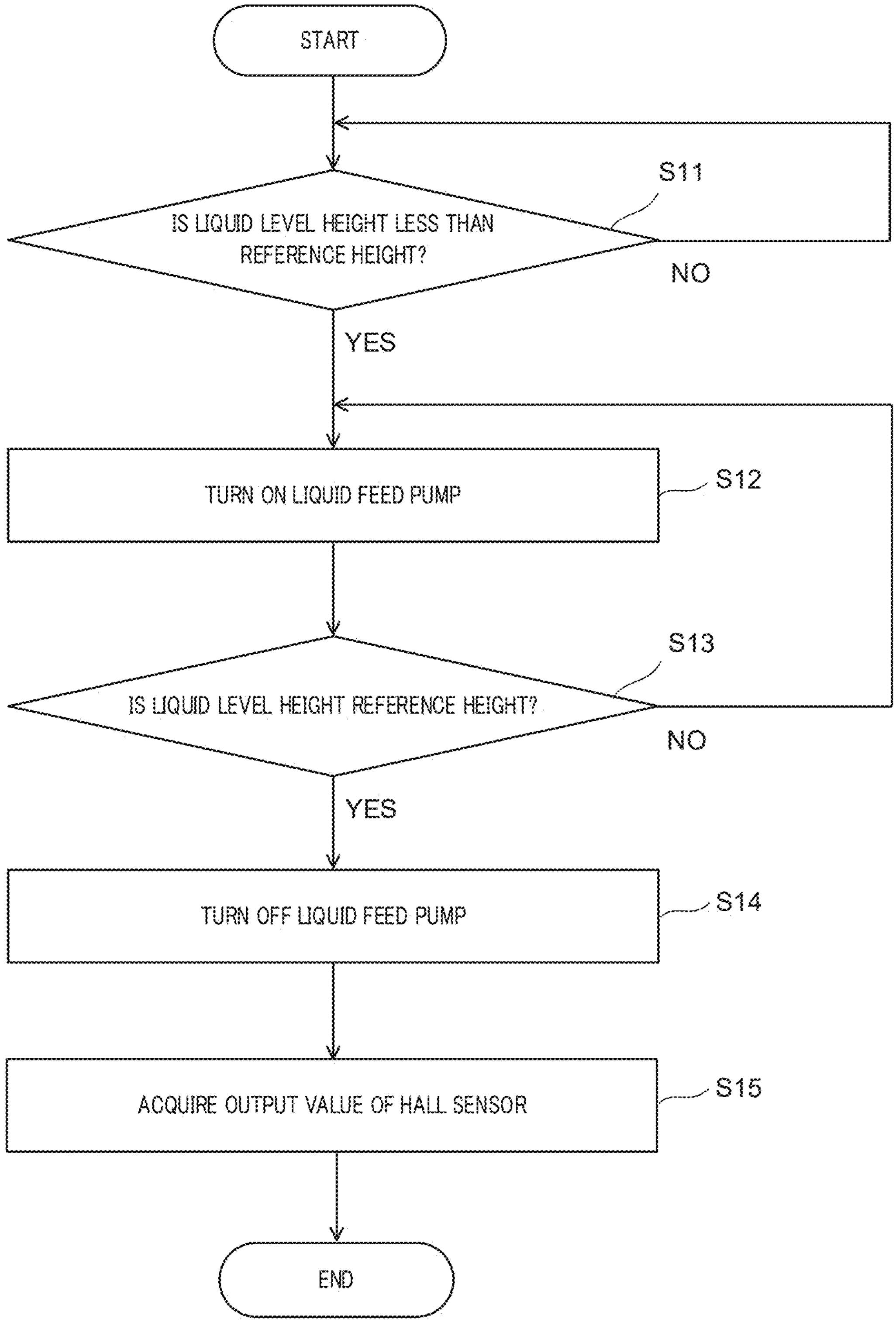

START
IS LIQUID LEVEL HEIGHT LESS THAN REFERENCE HEIGHT?
S11
NO
YES
TURN ON LIQUID FEED PUMP
S12
IS LIQUID LEVEL HEIGHT REFERENCE HEIGHT?
S13
NO
YES
TURN OFF LIQUID FEED PUMP
S14
ACQUIRE OUTPUT VALUE OF HALL SENSOR
S15
END

FIG. 3

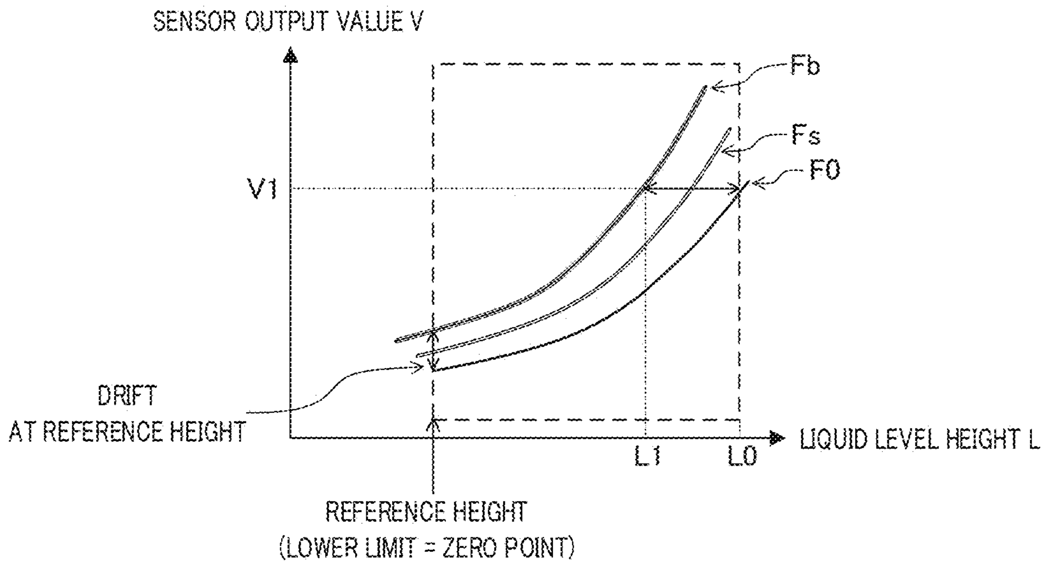

SENSOR OUTPUT VALUE V
Fb
Fs
F0
V1
DRIFT
AT REFERENCE HEIGHT
L1
L0
LIQUID LEVEL HEIGHT L
REFERENCE HEIGHT
(LOWER LIMIT = ZERO POINT)

FIG. 4A

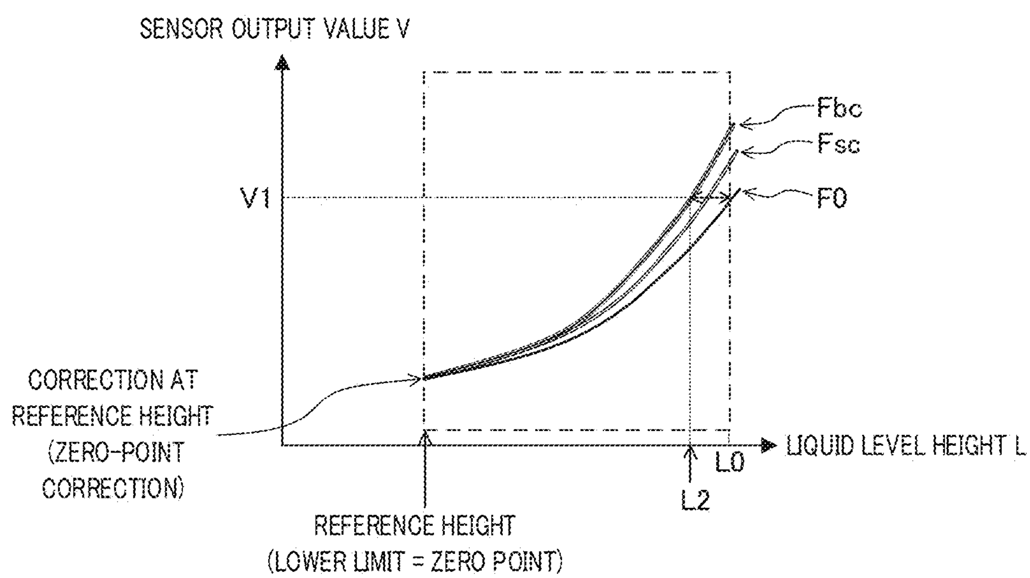

SENSOR OUTPUT VALUE V
Fbc
Fsc
F0
V1
CORRECTION AT
REFERENCE HEIGHT
(ZERO-POINT
CORRECTION)
L0
L2
LIQUID LEVEL HEIGHT L
REFERENCE HEIGHT
(LOWER LIMIT = ZERO POINT)

FIG. 4B

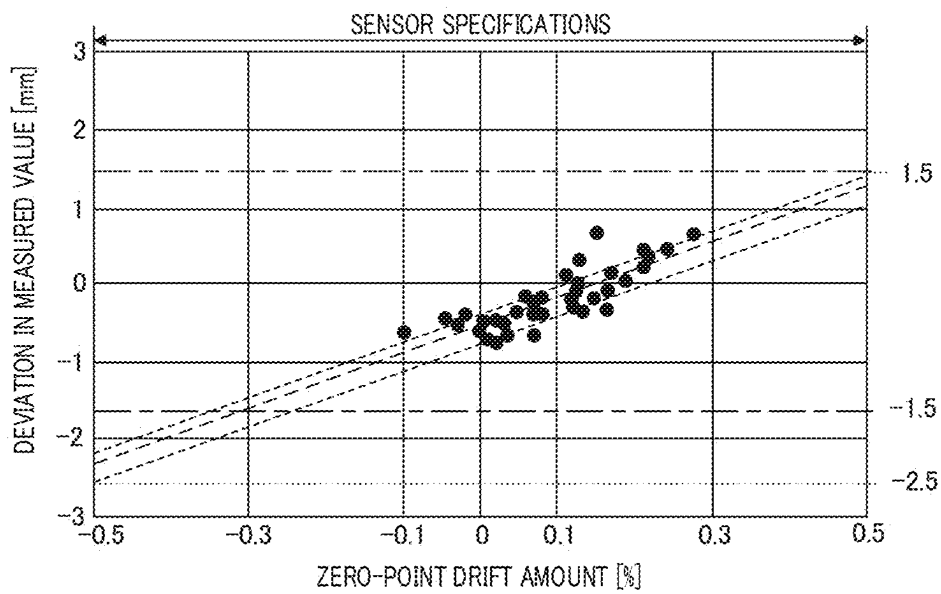

SENSOR SPECIFICATIONS
DEVIATION IN MEASURED VALUE [mm]
3
2
1
0
−1
−2
−3
1.5
−1.5
−2.5
−0.5
−0.3
−0.1
0
0.1
0.3
0.5
ZERO-POINT DRIFT AMOUNT [%]

FIG. 5A

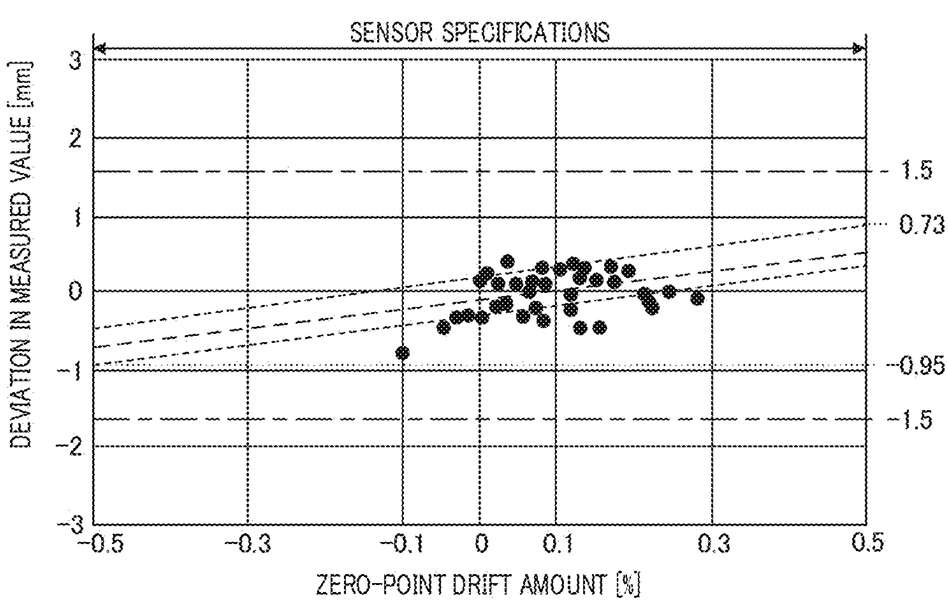

SENSOR SPECIFICATIONS
DEVIATION IN MEASURED VALUE [mm]
3
2
1
0
−1
−2
−3
1.5
0.73
−0.95
−1.5
−0.5
−0.3
−0.1
0
0.1
0.3
0.5
ZERO-POINT DRIFT AMOUNT [%]

FIG. 5B

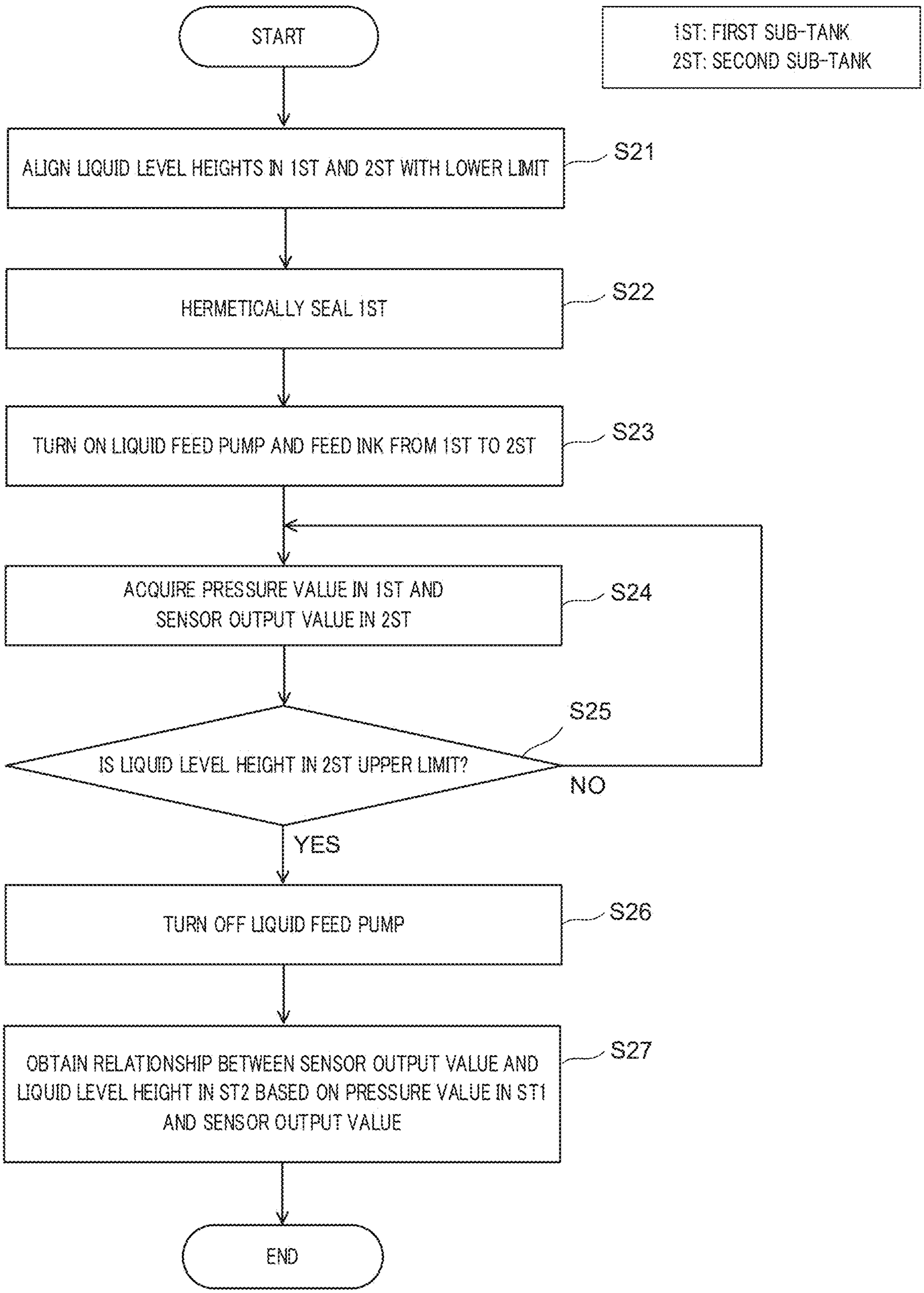

1ST: FIRST SUB-TANK
2ST: SECOND SUB-TANK
START
ALIGN LIQUID LEVEL HEIGHTS IN 1ST AND 2ST WITH LOWER LIMIT
S21
HERMETICALLY SEAL 1ST
S22
TURN ON LIQUID FEED PUMP AND FEED INK FROM 1ST TO 2ST
S23
ACQUIRE PRESSURE VALUE IN 1ST AND SENSOR OUTPUT VALUE IN 2ST
S24
IS LIQUID LEVEL HEIGHT IN 2ST UPPER LIMIT?
S25
NO
YES
TURN OFF LIQUID FEED PUMP
S26
OBTAIN RELATIONSHIP BETWEEN SENSOR OUTPUT VALUE AND LIQUID LEVEL HEIGHT IN ST2 BASED ON PRESSURE VALUE IN ST1 AND SENSOR OUTPUT VALUE
S27
END

FIG. 6

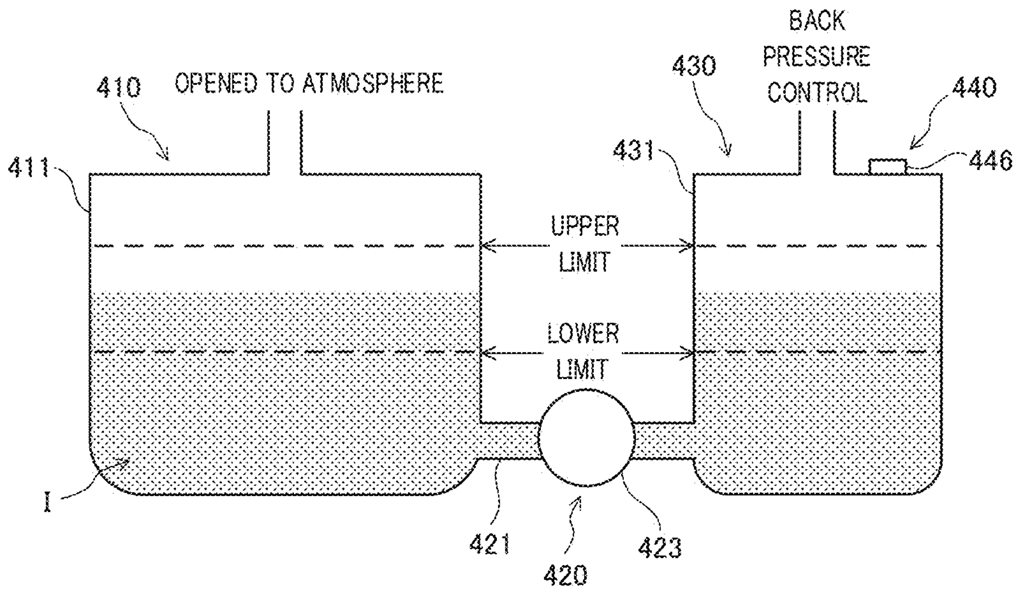

410
OPENED TO ATMOSPHERE
430
BACK
PRESSURE
CONTROL
440
411
431
446
UPPER
LIMIT
LOWER
LIMIT
I
421
420
423

FIG. 7A

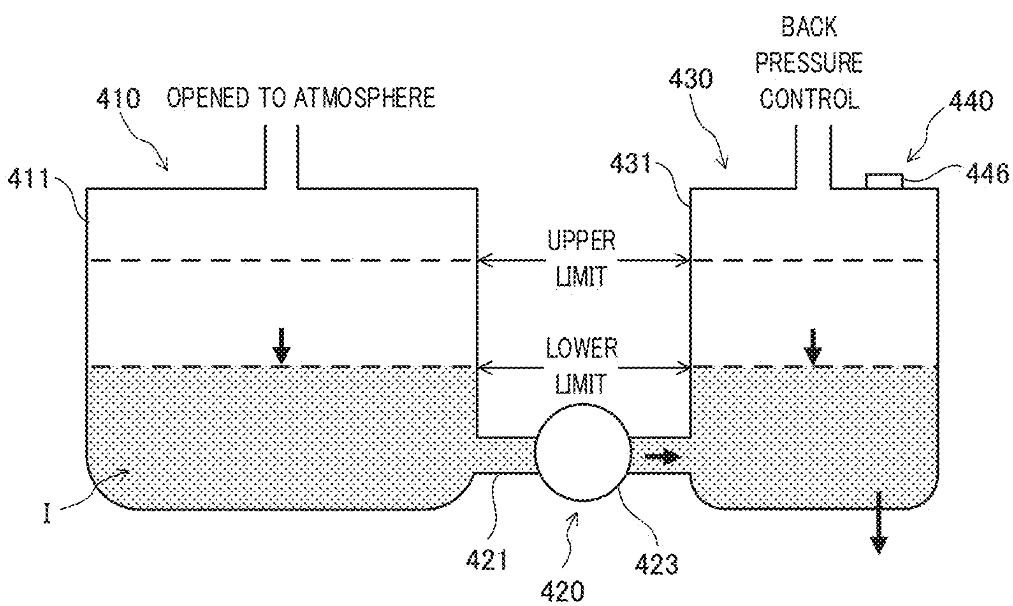

410
OPENED TO ATMOSPHERE
430
BACK
PRESSURE
CONTROL
440
411
431
446
UPPER
LIMIT
LOWER
LIMIT
I
421
420
423

FIG. 7B

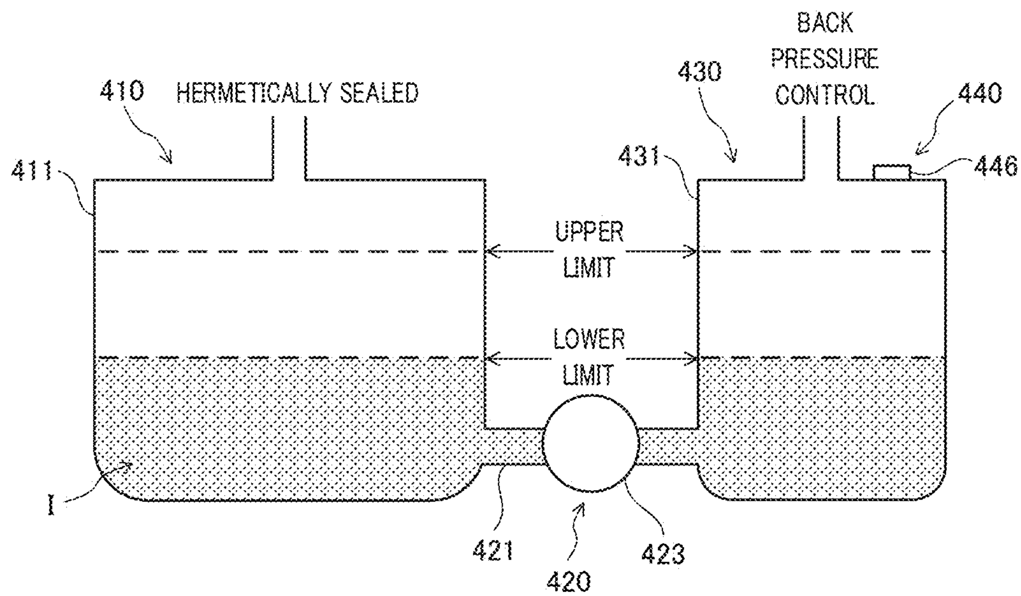

410
HERMETICALLY SEALED
BACK
PRESSURE
CONTROL
430
440
411
431
446
UPPER
LIMIT
LOWER
LIMIT
I
421
420
423

FIG. 7C

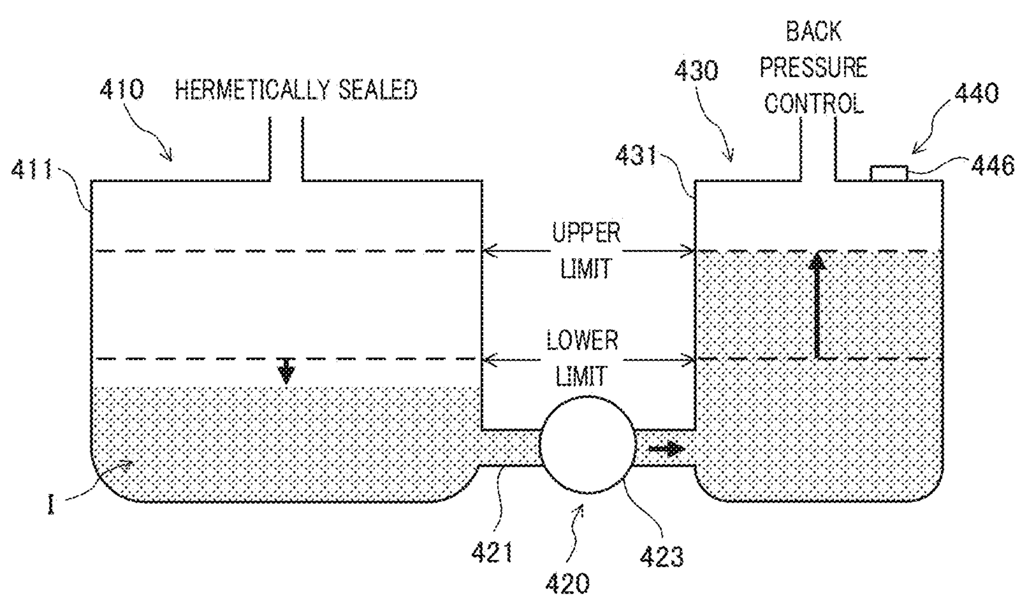

410
HERMETICALLY SEALED
BACK
PRESSURE
CONTROL
430
440
411
431
446
UPPER
LIMIT
LOWER
LIMIT
I
421
420
423

FIG. 7D

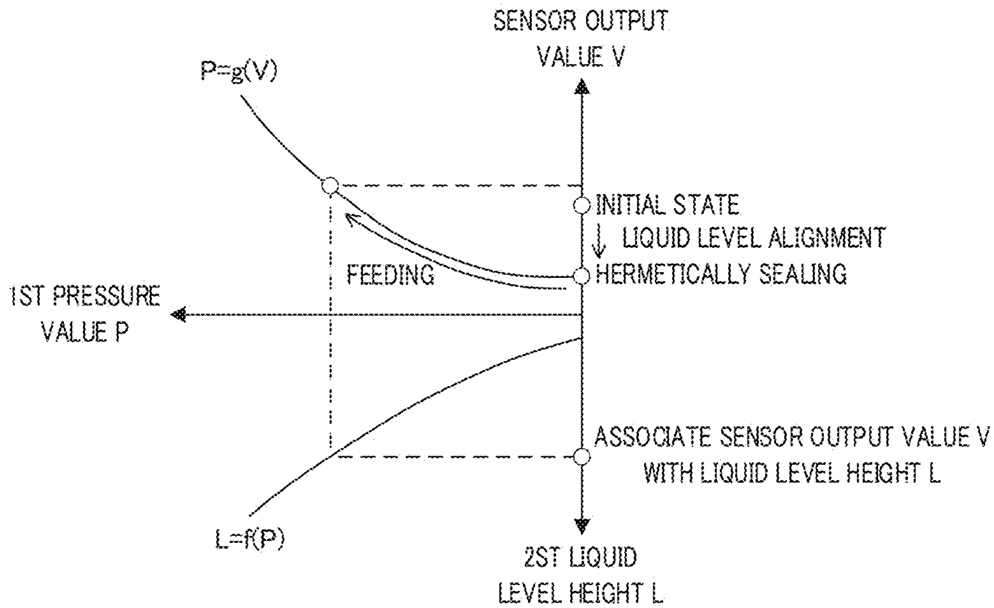

SENSOR OUTPUT
VALUE V
P=g(V)
INITIAL STATE
LIQUID LEVEL ALIGNMENT
HERMETICALLY SEALING
FEEDING
1ST PRESSURE
VALUE P
ASSOCIATE SENSOR OUTPUT VALUE V
WITH LIQUID LEVEL HEIGHT L
L=f(P)
2ST LIQUID
LEVEL HEIGHT L

FIG. 8A

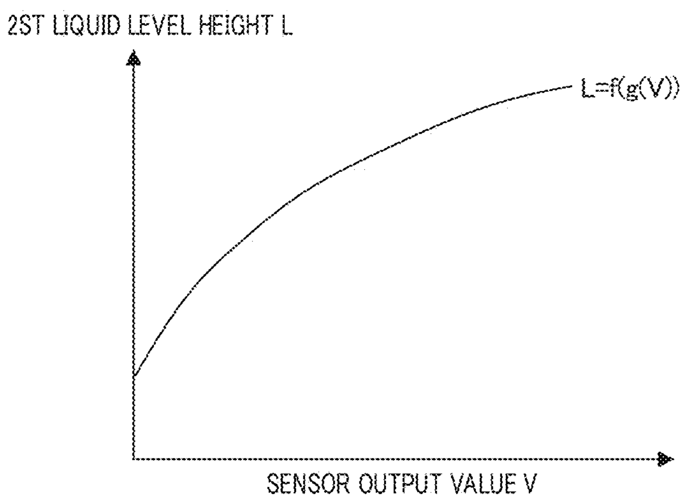

2ST LIQUID LEVEL HEIGHT L
L=f(g(V))
SENSOR OUTPUT VALUE V

FIG. 8B

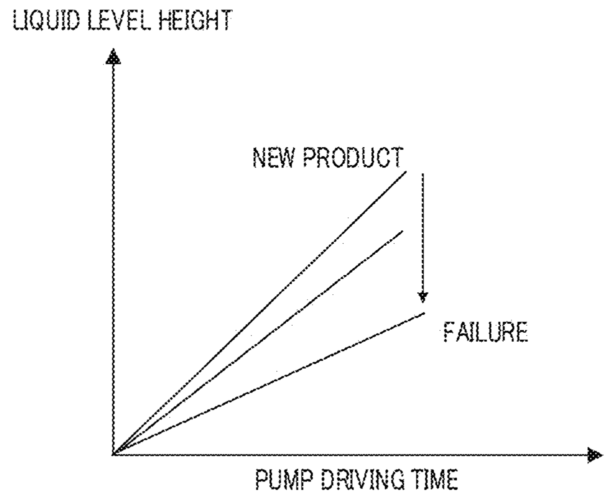

LIQUID LEVEL HEIGHT
NEW PRODUCT
FAILURE
PUMP DRIVING TIME

FIG. 9

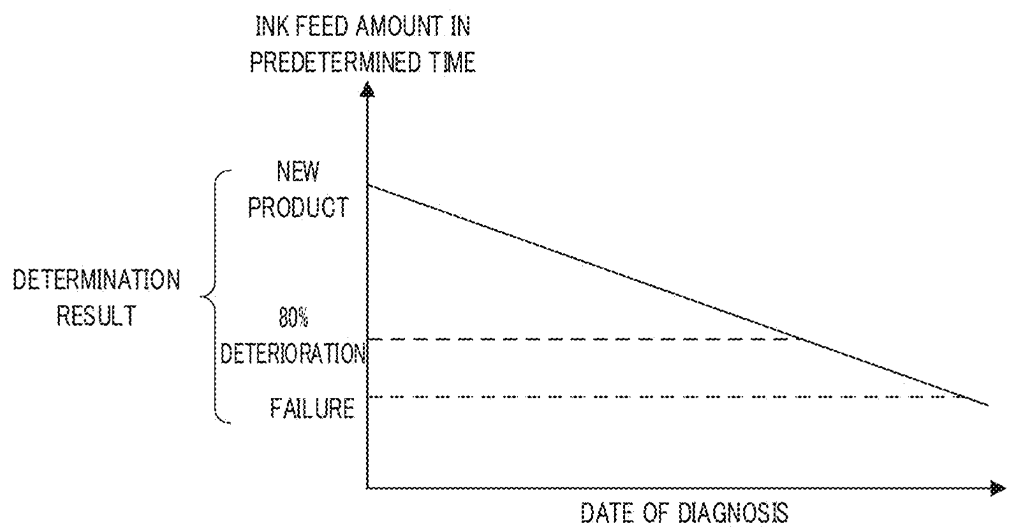

INK FEED AMOUNT IN
PREDETERMINED TIME
DETERMINATION
RESULT
NEW
PRODUCT
80%
DETERIORATION
FAILURE
DATE OF DIAGNOSIS

FIG. 10

INK LIQUID LEVEL HEIGHT MEASURING APPARATUS AND IMAGE FORMING APPARATUS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The entire disclosure of Japanese Patent Application No. 2024-025674, filed on Feb. 22, 2024, is incorporated herein by reference in its entirety.

BACKGROUND

Technological Field

[0002] The present invention relates to an ink liquid level height measuring apparatus and an image forming apparatus.

Description of Related Art

[0003] In an inkjet-type image forming apparatus, a tank that stores ink to be supplied to an inkjet head is provided with an ink liquid level height measuring apparatus that measures the liquid level height of the ink in the tank.

[0004] In the invention disclosed in Japanese Patent Publication Laid-Open No. 2021-783, an ink liquid level height measuring apparatus is configured to measure the upper and lower limits of the liquid level in order to detect an abnormality of a component of an ink supply apparatus for supplying ink to a tank. The liquid level height measured by the ink liquid level height measuring apparatus is not a continuous value but a discrete value between the upper and lower limits, and the liquid level height between the upper and lower limit is unknown. For this reason, there has been a problem in that it is difficult to detect an abnormality of a component of the ink supply apparatus at a value other than the upper and lower limits, or that even when an abnormality can be detected, the detection accuracy is insufficient.

[0005] In the invention disclosed in Japanese Patent Publication Laid-Open No. 2020-34485, an ink liquid level height measuring apparatus is disclosed which is capable of continuously measuring changes in the liquid level height of ink in a tank by using a capacitive sensor. However, the capacitive sensor has a problem in that sensitivity cannot be obtained depending on the physical properties of ink.

[0006] The inventions disclosed in Japanese Patent Publication Laid-Open No. 2021-783 and Japanese Patent Publication Laid-Open No. 2020-34485 involve the above-described problems. In this respect, the present inventor has considered a method using a Hall element as a method of continuously measuring changes in the liquid level height of ink without depending on the physical properties of ink. In order to accurately detect an abnormality of a component of an ink supply apparatus, it is necessary to measure minor changes in the liquid level height, but the Hall element has a characteristic that an output value thereof changes over time (drifts).

SUMMARY

[0007] An object of the present invention is to provide an ink liquid level height measuring apparatus and an image forming apparatus each capable of accurately measuring the liquid level height of ink by using a Hall element even when there is a change over time.

[0008] In order to achieve at least one of the above-described objects, an ink liquid level height measuring apparatus reflecting one aspect of the present invention is an ink liquid level height measuring apparatus for measuring a liquid level height of ink in a container and includes: a float that includes a magnet and moves in association with a change in the liquid level height; a Hall element that detects, as a continuous value, a change in magnetic flux density due to the magnet that moves together with the float; and at least one hardware processor, and the at least one hardware processor adjusts the liquid level height to a reference height, and acquires an output value of the Hall element, when the liquid level height is the reference height, and corrects the liquid level height based on the output value.

[0009] In order to achieve at least one of the above-described objects, an image forming apparatus reflecting one aspect of the present invention includes: a container that stores ink; an image former that forms an image by using the ink supplied from the container; and the ink liquid level height measuring apparatus described above.

BRIEF DESCRIPTION OF DRAWINGS

[0010] The advantages and features provided by one or more embodiments of the invention will become more fully understood from the detailed description given hereinbelow and the appended drawings which are given by way of illustration only, and thus are not intended as a definition of the limits of the present invention:

[0011] FIG. **1** is a schematic diagram illustrating examples of an image forming apparatus and an ink liquid level height measuring apparatus according to an embodiment of the present invention;

[0012] FIG. **2** is a block diagram illustrating a main part of a control system of the image forming apparatus illustrated in FIG. **1**;

[0013] FIG. **3** is a flowchart illustrating zero-point correction of a Hall sensor of the ink liquid level height measuring apparatus;

[0014] FIG. **4**A is a graph illustrating the relationship between the liquid level height before zero-point correction and the sensor output value;

[0015] FIG. **4**B is a graph illustrating the relationship between the liquid level height after zero-point correction and the sensor output value;

[0016] FIG. **5**A illustrates the relationship between the zero-point drift amount and the deviation in the measured value in the case of no zero-point correction;

[0017] FIG. **5**B illustrates the relationship between the zero-point drift amount and the deviation in the measured value in the case of zero-point correction;

[0018] FIG. **6** is a flowchart illustrating entire-region correction of the Hall sensor of the ink liquid level height measuring apparatus;

[0019] FIG. **7**A illustrates initial states of a first sub-tank and a second sub-tank in the entire-region correction illustrated in FIG. **6**;

[0020] FIG. **7**B illustrates a state in which liquid level alignment is performed in the first sub-tank and the second sub-tank in the entire-region correction illustrated in FIG. **6**;

[0021] FIG. **7**C illustrates a state in which the first sub-tank is hermetically sealed in the entire-region correction illustrated in FIG. **6**;

[0022] FIG. **7**D illustrates a state in which ink is being fed from the first sub-tank to the second sub-tank in the entire-region correction illustrated in FIG. **6**;

[0023] FIG. **8**A illustrates the relationships of the liquid level height and the sensor output value in the second sub-tank to the pressure value in the first sub-tank in the entire-region correction illustrated in FIG. **6**;

[0024] FIG. **8**B is a graph illustrating the relationship between the sensor output value and the liquid level height, which is obtained from the relationships illustrated in FIG. **8**A;

[0025] FIG. **9** is a graph illustrating the relationship between the pump driving time and the liquid level height from a state in which an ink supply section is a new product to a state in which the ink supply section is determined to have a failure; and

[0026] FIG. **10** is a graph illustrating a determination result obtained by making determination based on a change over time in the liquid feed amount of ink supplied for a predetermined time.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] Hereinafter, one or more embodiments of the present invention will be described with reference to the drawings. However, the scope of the invention is not limited to the disclosed embodiments.

[0028] Hereinafter, an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[Image Forming Apparatus]

[0029] FIG. **1** is a schematic diagram illustrating examples of an inkjet printer **100** (the image forming apparatus in the present invention) and an ink liquid level height measuring apparatus **440** according to the present embodiment. FIG. **2** is a block diagram illustrating a main part of a control system of the inkjet printer **100** illustrated in FIG. **1**.

[0030] As illustrated in FIG. **2**, the inkjet printer **100** includes a conveyance section **10**, a supply section **20**, a discharge section **30**, an ink supply section **40**, a head module **50** (the image forming section (image former) in the present invention), an operation display section **70**, an input/output interface **80**, a control section **90**, and the like.

[0031] The conveyance section **10** conveys a recording medium M (see FIG. **1**). The conveyance section **10** is constituted by, for example, a conveyance belt, a conveyance drum, and the like. The recording medium M supplied from the supply section **20** is conveyed to the head module **50** by a conveyance operation of the conveyance section **10**. Thereafter, the recording medium M on which an image has been formed by the head module **50** is conveyed to the discharge section **30** by a conveyance operation of the conveyance section **10**.

[0032] As the recording medium M, it is possible to use various media on which ink ejected from an inkjet head (illustration is omitted) of the head module **50** can be fixed. The recording medium M is, for example, a medium such as sheet-like paper, cloth (fabric), or resin. The recording medium M is not limited to a sheet-like medium, and may be a medium such as roll-shaped paper, cloth, or resin.

[0033] The supply section **20** stores the recording medium M and supplies the recording medium M to the conveyance section **10**. The supply section **20** includes, for example, a storage section that stores the recording medium M, and includes a belt, a roller, and the like that convey the recording medium M to the conveyance section **10**.

[0034] The discharge section **30** stores the recording medium M discharged from the conveyance section **10**. The discharge section **30** includes, for example, a belt, a roller, and the like that convey the recording medium M from the conveyance section **10**, and includes a storage section that stores the recording medium M.

[0035] The ink supply section **40** is an apparatus that supplies ink I to the head module **50**. The ink supply section **40** includes a first sub-tank section **410**, a liquid feed section **420**, a second sub-tank section **430**, the ink liquid level height measuring apparatus **440**, and the like.

[0036] The first sub-tank section **410** stores the ink I to be fed to the second sub-tank section **430**. The first sub-tank section **410** includes a first sub-tank **411** (the supply source container in the present invention) or the like that stores the ink I. The first sub-tank **411** is connected to a main tank (illustration is omitted) that stores the ink I via a liquid feed path (illustration is omitted). The ink I stored in the main tank is fed to the first sub-tank **411** by using a pump or the like (illustration is omitted).

[0037] In addition, the first sub-tank section **410** includes, as a reference height detection section D**1** that detects whether the liquid level height of the ink I in the first sub-tank **411** is the reference height, a float **412**, a guide section **413**, a magnet **414**, and a reed switch **416**.

[0038] The float **412** is a floating body that floats on the liquid level of the ink I. In a plan view, a through-hole into which the guide section **413** is inserted is formed in a center portion of the float **412**.

[0039] The guide section **413** is a rod-shaped member that includes an end part supported by a side of a top plate of the first sub-tank **411** and extends downward. The guide section **413** is inserted into the through-hole of the float **412**. The guide section **413** regulates the float **412** such that the float **412** does not tilt, and guides the float **412** such that the float **412** is movable in the vertical direction that is a liquid level height direction, that is, the float **412** is movable up and down. Accordingly, the float **412** moves up and down along the guide section **413** as the liquid level of the ink I moves up and down.

[0040] Note that the guide section **413** may be a rod-shaped member that extends upward and includes an end part supported by a side of a bottom plate of the first sub-tank **411** as long as the guide section **413** can regulate the float **412** such that the float **412** does not tilt, and can guide the float **412** such that the float **412** is movable up and down. Further, as long as the float **412** can be regulated so as not to tilt, the guide section **413** and the through-hole of the float **412** may be omitted. For example, when the outer peripheral side surface of the float **412** is formed so as to be in proximity to the inner wall of the first sub-tank **411**, the inner wall of the first sub-tank **411** functions as a guide section that regulates the float **412** such that the float **412** does not tilt, and guides the float **412** such that the float **412** is movable up and down.

[0041] The magnet **414** which is a permanent magnet is disposed inside the float **412**. The magnet **414** is, for example, an annular body having a rectangular cross section. The magnet **414** is disposed, inside the float **412**, in a position in which the float **412** floating on the liquid level of the ink I is balanced so as not to tilt.

[0042] The reed switch **416** is a sensor that detects whether the liquid level height of the ink I in the first sub-tank **411** is the reference height, and is disposed at the reference height. The reed switch **416** is a magnetic switch that is turned on or off by a magnetic force, and is turned on by the magnetic force of the magnet **414** in the float **412** when the liquid level height of the ink I is the reference height, and is turned off otherwise.

[0043] In this manner, in the first sub-tank section **410**, the reference height detection section D**1** detects whether the liquid level height of the ink I in the first sub-tank **411** is the reference height. Note that, although a float-type level switch is exemplified here as the reference height detection section D**1**, another type, for example, a capacitive level switch or the like may also be used.

[0044] In addition, the first sub-tank unit **410** includes a valve **417**, which seals the upper space of the first sub-tank **411** and releases the upper space to the atmosphere, and a pressure sensor **418**, which detects the pressure in the first sub-tank **411**. The valve **417** and the pressure sensor **418** are used for correction to be described later.

[0045] The liquid feed section **420** feeds the ink I stored in the first sub-tank **411** to the second sub-tank section **430** (a second sub-tank **431** to be described later). The liquid feed section **420** includes a tank supply path **421** (the ink supply path in the present invention), a deaeration module **422**, a liquid feed pump **423**, a liquid feed valve **424**, and the like.

[0046] The tank supply path **421** connects the first sub-tank **411** to the second sub-tank **431**, and serves as a flow path from the first sub-tank **411** to the second sub-tank **431**. The deaeration module **422** includes a filter **425**. The deaeration module **422** deaerates the ink I to be fed, and filters the ink I with the filter **425**.

[0047] The liquid feed pump **423** is a pump that feeds the ink I from the first sub-tank **411** to the second sub-tank **431**. The liquid feed valve **424** opens and closes the tank supply path **421**. For example, the liquid feed valve **424** opens when the ink I is fed from the first sub-tank **411** to the second sub-tank **431**, and the liquid feed valve **424** closes when the ink I is fed from the second sub-tank **431** to the head module **50**.

[0048] The second sub-tank section **430** stores the ink I to be fed to the head module **50**. The second sub-tank section **430** includes the second sub-tank **431** (the container in the present invention) that stores the ink I, a pneumatic pump **432**, a head supply path **433**, the ink liquid level height measuring apparatus **440**, and the like.

[0049] The second sub-tank **431** is connected to the first sub-tank **411** via the tank supply path **421**, and is connected to the head module **50** via the head supply path **433**.

[0050] The pneumatic pump **432** is connected to an upper portion of the second sub-tank **431**, and controls the pressure in an upper space of the second sub-tank **431** to a desired pressure. The head supply path **433** connects a lower portion of the second sub-tank **431** to the head module **50**, and supplies the ink I from the second sub-tank **431** to the head module **50**.

[0051] The second sub-tank section **430** described above is provided with the ink liquid level height measuring apparatus **440** that measures the liquid level height of the ink I in the second sub-tank **431**. The ink liquid level height measuring apparatus **440** may also be provided in the first sub-tank section **410** to measure the liquid level height of the ink I in the first sub-tank **411**. The ink liquid level height measuring apparatus **440** will be described later.

[0052] The head module **50** includes apparatuses and members necessary for image formation, such as an inkjet head. The head module **50** ejects, from a nozzle of the inkjet head, the ink I supplied from the ink supply section **40** (the second sub-tank section **430**) to form an image on the recording medium M.

[0053] Note that, in FIG. **1**, in order to simplify the drawing, the ink supply section **40** and the head module **50** for one color are illustrated, but ink supply sections **40** and head modules **50** are disposed according to the number of colors to be used. For example, in a case where four colors of yellow (Y), magenta (M), cyan (C), and black (K) are used, ink supply sections **40** and head modules **50** for the four colors are disposed.

[0054] The operation display section **70** is, for example, a flat panel display, such as a liquid crystal flat panel display or an organic electro luminescence (EL) flat panel display, with a touch screen. The operation display section **70** displays an operation menu for the user, information on image data, various states of the inkjet printer **100**, and the like. In addition, the operation display section **70** includes a plurality of keys, and receives various input operations of the user.

[0055] The input/output interface **80** mediates transmission and reception of data between an external apparatus **200** and the control section **90**. Input/output interface **80** is constituted by, for example, one of various serial interfaces and various parallel interfaces, or a combination thereof.

[0056] The external apparatus **200** is, for example, a personal computer, a facsimile machine, or the like, and supplies a print job, image data, and the like, to the control section **90** via the input/output interface **80**.

[0057] The control section **90** includes a central processing unit (CPU) **91**, a random access memory (RAM) **92**, a read only memory (ROM) **93**, a storage section **94**, and the like as illustrated in FIG. **2**.

[0058] The CPU **91** reads various control programs and setting data stored in the ROM **93**, stores the read programs and setting data in the RAM **92**, and executes the programs to carry out various pieces of calculation processing. For example, the control section **90** generates, based on image data received from the input/output interface **80**, a driving signal for an image to be formed, and outputs the driving signal to the inkjet head.

[0059] The RAM **92** provides a working memory space for the CPU **91** and stores temporary data. Note that, the RAM **92** may include a non-volatile memory.

[0060] The ROM **93** stores the various control programs to be executed by the CPU **91**, the setting data, and the like. Note that, a rewritable non-volatile memory such as an electrically erasable programmable read only memory (EEPROM) or a flash memory may also be used instead of the ROM **93**.

[0061] The storage section **94** stores a print job and image data associated with a print job, which are inputted from the external apparatus **200** via the input/output interface **80**. As the storage section **94**, for example, a hard disk drive (HDD), a solid state drive (SSD) or the like is used, and a dynamic random access memory (DRAM) or the like may also be used in combination.

[0062] The conveyance section **10**, the supply section **20**, the discharge section **30**, the ink supply section **40**, the head module **50**, the operation display section **70**, the input/output interface **80**, and the like are connected to the control section **90**. The control section **90** integrally controls the entire operation of the inkjet printer **100**. The conveyance section **10**, the supply section **20**, the discharge section **30**, the ink supply section **40**, the head module **50**, the operation display section **70**, the input/output interface **80**, and the like are controlled by the control section **90** to execute predetermined processing.

[0063] With the above-described configuration, the inkjet printer **100** supplies the recording medium M from the supply section **20** to the conveyance section **10**, forms an image on the recording medium M, which is conveyed by the conveyance section **10**, with the head module **50**, and conveys the recording medium M, on which the image has been formed, to the discharge section **30**.

[Ink Liquid Level Height Measuring Apparatus]

[0064] As illustrated in FIG. **1**, the second sub-tank section **430** of the ink supply section **40** is provided with the ink liquid level height measuring apparatus **440**. The ink liquid level height measuring apparatus **440** includes a float **441**, a guide section **442**, a magnet **443**, a reed switch **445**, a Hall sensor **446**, a magnetic body **447**, and a measurement control section **448**.

[0065] The second sub-tank section **430** includes, as a reference height detection section D**2** that detects whether the liquid level height of the ink I in the second sub-tank **431** is the reference height, the float **441**, the guide section **442**, the magnet **443**, and the reed switch **445**.

[0066] The float **441**, the guide section **442**, the magnet **443**, and the reed switch **445** have the same configurations as those of the float **412**, the guide section **413**, the magnet **414**, and the reed switch **416** in the reference height detection section D**1**, and overlapping descriptions thereof will be omitted here. Here, the guide section **442** regulates the movement of the float **441** in the horizontal direction such that the position of the magnet **443** with respect to the Hall sensor **446** does not change in the horizontal direction along the liquid level.

[0067] In the second sub-tank section **430**, the reference height detection section D**2** detects whether the liquid level height of the ink I in the second sub-tank **431** is the reference height. Note that, although a float-type level switch is also exemplified here as the reference height detection section D**2**, another type, for example, a capacitive level switch or the like may be also used.

[0068] The Hall sensor **446** is a magnetic sensor using a Hall element that detects the strength of a magnetic field. The Hall sensor **446** may include an operational amplifier circuit or the like.

[0069] The Hall sensor **446** is disposed in a position in which the Hall sensor **446** can measure a change in magnetic flux density due to the magnet **443** that moves up and down together with the float **441**, preferably, in a position in which the magnetic flux density increases. For example, the Hall sensor **446** is fixed to a side of a top plate of the second sub-tank **431** together with the magnetic body **447**, and is disposed in a position in which the Hall sensor **446** and the magnetic body **447** face the magnet **443** in the vertical direction.

[0070] With the above-described configuration, the Hall sensor **446** detects, as a continuous value, a change in the magnetic flux density, which is a change in the physical quantity associated with a change in the liquid level height, and outputs the sensor output value.

[0071] The magnetic body **447** is a member made of a material having a high magnetic permeability such as iron, for example, SS400 which is a rolled steel material for general structure. Usually, the magnetic field is distorted so as to be directed toward a material a having high magnetic permeability, for example, a magnetic body, and the magnetic flux is concentrated. Accordingly, by disposing the magnetic body **447** in proximity to the Hall sensor **446**, the magnetic flux from the magnet **443** can be concentrated in the position of the Hall sensor **446** to increase the magnetic flux density.

[0072] In particular, in a case where the magnet **443** is disposed on the side beneath the Hall sensor **446**, the magnetic flux density in the sensor position is larger when the magnetic body **447** is disposed in proximity to and on the upper side of a housing of the Hall sensor **446**. Accordingly, in FIG. **1**, as an example, the magnetic body **447** is disposed on the upper surface of the housing of the Hall sensor **446**. Further, as the size (width, thickness, and depth) of the magnetic body **447** increases, the magnetic flux density in the sensor position increases.

[0073] Since the magnetic body **447** is disposed on and in proximity to the upper side of the housing of the Hall sensor **446**, the Hall sensor **446** detects, as a continuous value, the liquid level height of the ink I, that is, the magnetic flux density that changes as the float **441** moves up and down, with high sensitivity, and outputs an output value corresponding to the liquid level height. As a result, the ink liquid level height measuring apparatus **440** can continuously measure the liquid level height of the ink I with high accuracy.

[0074] Note that, the Hall sensor **446** and the magnetic body **447** may be fixed to a side of a bottom plate of the second sub-tank **431**, and may be disposed in a position in which the Hall sensor **446** and the magnetic body **447** face the magnet **443** in the vertical direction. In that case, the magnetic body **447** is desirably disposed on the lower surface of the housing of the Hall sensor **446**.

[0075] Although illustration is omitted, the measurement control section **448** includes a CPU, a RAM, a ROM, a storage section, and the like. The CPU reads a program and data stored in the ROM, stores the read program and data in the RAM, and executes the program to carry out various pieces of processing. For example, the measurement control section **448** obtains the liquid level height based on the output value from the Hall sensor **446**.

[0076] Incidentally, as described above, the Hall element used in the Hall sensor **446** has a characteristic that an output value thereof changes over time (drifts).

[0077] Accordingly, in the present embodiment, in order to accurately measure the liquid level height of the ink I by the Hall sensor **446** including the Hall element even when there is a change over time, the ink liquid level height measuring apparatus **440** has a configuration to be described below.

[0078] Specifically, the ink liquid level height measuring apparatus **440** includes an adjustment section, which adjusts the liquid level height of the ink I to the reference height, and a correction section (corrector), which corrects the liquid level height based on the output of the Hall sensor **446** when the liquid level height is the reference height. Here, the measurement control section **448** may be constituted by at least one hardware processor, in which case the at least one hardware processor functions as the adjustment section and the correction section, and also functions as an acquisition section (acquirer) and a determination section both of which will be described later. In addition, the adjustment section and the correction section, as well as the acquisition section and the determination section both of which will be described later, may be provided as functions of the measurement control section **448**, for example, may be provided as programs that are executed by the measurement control section **448**.

[Correction 1]

[0079] Correction 1 (zero-point correction), which is an example of correction using the adjustment section and the correction section, will be described with reference to FIG. **3** together with FIG. **1**. FIG. **3** is a flowchart illustrating zero-point correction of the Hall sensor **446** of the ink liquid level measuring apparatus **440**. Further, since the correction is performed here at the zero point, that is, the lower limit, the reference height in the reference height detection section D**2** is set to the lower limit of the second sub-tank **431**.

[0080] In the inkjet printer **100**, when an image is formed on the recording medium M by a print job, the ink I in the second sub-tank **431** is consumed, and the liquid level height of the ink I in the second sub-tank **431** becomes less than the reference height (the lower limit). As seen above, the zero-point correction to be described below is executed when the liquid level height of the ink I in the second sub-tank **431** becomes less than the reference height (the lower limit) during image formation by a print job.

(Step S**11**)

[0081] The measurement control section **448** (the adjustment section) confirms whether the liquid level height of the ink I in the second sub-tank **431** is less than the reference height by using the reference height detection section D**2**. In a case where the liquid level height of the ink I in the second sub-tank **431** is less than the reference height (YES), the processing proceeds to step S**12**, and in a case where the liquid level height of the ink I in the second sub-tank **431** is not less than the reference height (NO), step S**11** is repeated.

(Step S**12**)

[0082] The measurement control section **448** (the adjustment section) turns on the liquid feed pump **423** (also opens the liquid feed valve **424**) and feeds the ink I from the first sub-tank **411** to the second sub-tank **431**.

(Step S**13**)

[0083] The measurement control section **448** (the adjustment section) confirms whether the liquid level height of the ink I in the second sub-tank **431** is the reference height by using the reference height detection section D**2**. In a case where the liquid level height of the ink I in the second sub-tank **431** is the reference height (YES), the processing proceeds to step S**14**, and in a case where the liquid level height of the ink I in the second sub-tank **431** is not less than the reference height (NO), steps S**12** and S**13** are repeated.

(Step S**14**)

[0084] The measurement control section **448** (the adjustment section) turns off the liquid feed pump **423** (also closes the liquid feed valve **424**) to stop feeding the ink I from the first sub-tank **411** to the second sub-tank **431**. That is, when the liquid level height of the ink I in the second sub-tank **431** reaches the reference height, the supply of the ink I is stopped.

(Step S**15**)

[0085] The measurement control section **448** (the correction section) acquires the output value of the Hall sensor **446** by using the ink liquid level height measuring apparatus **440** in a state in which the liquid level height of the ink I in the second sub-tank **431** is the reference height.

[0086] Then, the measurement control section **448** (the correction section) corrects the liquid level height as follows based on the output value of the Hall sensor **446** in a state in which the liquid level height of the ink I is the reference height.

[0087] As an example, it is assumed that a liquid level height L is obtained by the following calculation expression 1 based on an output value V of the Hall sensor **446**.

[1]

$$L = a \times V + b \qquad \text{(Calculation Expression 1)}$$

[0088] In the above calculation expression (linear expression), a coefficient a is a predetermined coefficient and is a slope of the calculation expression. In addition, a coefficient b is a predetermined coefficient and is an intercept of the calculation expression, but here, by correcting the coefficient b which is a constant term, the liquid level height of the ink I is accurately measured by using the Hall sensor **446** even when there is a change over time.

[0089] Specifically, at the start of use (initial stage) of the ink liquid level height measuring apparatus **440**, steps S**11** to S**15** described above are executed. Then, an output value V of the Hall sensor **446** in a state in which the liquid level height of the ink I in the second sub-tank **431** is the reference height is acquired as an initial value b0.

[0090] Thereafter, for example, when the liquid level height of the ink I in the second sub-tank **431** becomes less than the reference height, steps S**11** to S**15** described above are executed. Then, through steps S**11** to S**15** described above, the output value V of the Hall sensor **446** in a state in which the liquid level height of the ink I in the second sub-tank **431** is the reference height is acquired as a current value bt. A difference Δb between the initial value b0 and the current value bt is obtained as the correction value of the coefficient b, and the measurement control section **448** obtains the liquid level height L from the output value V of the Hall sensor **446** by using the following calculation expression 2 to which the difference Δb is added as the correction term of the coefficient b. According to the calculation expression 2, the liquid level height of the ink I can be accurately measured by using the Hall sensor **446** even when there is a change over time.

[2]

$$L = a \times V + (b + \Delta b) = a \times V + (b + b0 - bt) \quad \text{(Calculation Expression 2)}$$

[0091] Here, a case where the relationship between the liquid level height L and the output value V of the Hall sensor **446** is expressed by a linear expression has been exemplified. However, even in a case where the relationship therebetween is an N-th order expression of a second order or higher, the constant term (intercept) may be corrected in the same manner as the linear expression.

[0092] FIG. **4**A is a graph illustrating the relationship between the liquid level height L before zero-point correction and the output value V of the Hall sensor **446**. FIG. **4**B is a graph illustrating the relationship between the liquid level height L after zero-point correction and the output value V of the Hall sensor **446**. Note that the graphs of FIGS. **4**A and **4**B illustrate, as an example, a case where the relationship between the liquid level height L and the output value V of the Hall sensor **446** is expressed by a quadratic expression.

[0093] As described above, the Hall element used in the Hall sensor **446** has a characteristic that the output value thereof changes with time (drifts). For example, in FIG. **4**A, a graph **F0** is a graph illustrating the relationship between the liquid level height L and the output value V of the Hall sensor **446** at the start of use (initial stage) of the ink liquid level height measuring apparatus **440**. When the ink liquid level height measuring apparatus **440** is used for a long time, the output value V of the Hall sensor **446** with respect to the liquid level height L greatly drifts (see a graph Fb) or slightly drifts (see a graph Fs) from the graph **F0**, for example, due to a use environment or the like.

[0094] As illustrated in FIG. **4**A, when the output value V of the Hall sensor **446** drifts from the graph **F0** as the initial graph to the graph Fb or Fs, the liquid level height L is obtained as a liquid level height indicated in a range between L0 and L1, for example, even in the case of the same output value V=V1, and the accuracy of obtaining the liquid level height is not high.

[0095] Accordingly, in the present embodiment, the correction amount Δb for aligning (matching) the zero point of the graph Fb or Fs with (to) the zero point of the graph **F0** as the initial graph is obtained by using the current value bt acquired in steps S**11** to S**15** described above, and the zero-point correction is performed. As illustrated in FIG. **4**B, graphs Fbc and Fsc are obtained by performing the zero-point correction on the graphs Fb and Fs, respectively.

[0096] As a result, as illustrated in FIG. **4**B, the liquid level height L is obtained as a liquid level height indicated in a range between L0 and L1 which is narrower than the range between L0 and L2, for example, even in the case of the same output value V=V1, and the accuracy of obtaining the liquid level height improves.

[0097] Further, at a liquid level height close to the zero point (the lower limit), the drift amount in the graph Fbc or Fsc with respect to the graph **F0** is smaller as illustrated in FIG. **4**B, the accuracy of obtaining the liquid level height further improves.

[0098] Note that the term zero-point correction is used since the reference height here is the lower limit of the second sub-tank **431**, but the above-described correction may be performed by using a known height such as the middle position, the upper limit or the like of the second sub-tank **431** as the reference height.

[0099] FIGS. **5**A and **5**B are graphs in which the effects of the above-described zero-point correction are compared and verified. FIG. **5**A illustrates the relationship between the zero-point drift amount and the deviation in the measured value in the case of no zero-point correction. FIG. **5**B illustrates the relationship between the zero-point drift amount and the deviation in the measured value in the case of zero-point correction.

[0100] Here, a sensor in which the specifications of the zero-point drift due to a change over time are ±0.5% is used as the Hall sensor **446**, and the target value (target accuracy) of the deviation in the measured value is ±1.5 mm.

[0101] When the zero-point drift amount (%) and the deviation in the measured value (mm) were measured over a predetermined period of time by using the Hall sensor **446** as such with no zero-point correction, the results illustrated in FIG. **5**A were obtained. In this period of time, the zero-point drift amount did not drift up to 0.5% or close to −0.5%. When the obtained results are linearly approximated and the deviations in the measured values at 0.5% and −0.5% are predicted, the deviation is 1.5 mm at 0.5% and −2.5 mm at −0.5%, and thus, the target value of ±1.5 mm for the deviation in the measured values could not be satisfied.

[0102] In contrast, when the zero-point drift amount (%) and the deviation in the measured value (mm) were measured over a predetermined period of time with zero-point correction, the results illustrated in FIG. **5**B were obtained. Even in this period of time, the zero-point drift amount did not drift to 0.5% or close to −0.5%. When the obtained results are linearly approximated and the deviations in the measured values at 0.5% and −0.5% are predicted, the deviation is 0.73 mm at 0.5% and −0.95 mm at −0.5%, and thus, the target value of ±1.5 mm for the deviation in the measured value could be satisfied.

[0103] In this way, by performing the zero-point correction, the Hall sensor **446** can be used with the target accuracy of ±1.5 mm and the liquid level height can be accurately measured even when there is a change over time.

[0104] As described above, in the present embodiment, the inkjet printer **100** includes the ink liquid level height measuring apparatus **440**. In addition, the ink liquid level height measuring apparatus **440** includes the adjustment section, which adjusts the liquid level height of the ink I to the reference height, and the correction section, which corrects the liquid level height based on the output of the Hall sensor **446** when the liquid level height is the reference height.

[0105] According to the present embodiment configured as described above, the liquid level height is corrected based on the output of the Hall sensor **446** when the liquid level height is the reference height, and thus, it is possible to accurately measure the liquid level height of ink by using the Hall sensor **446** even when there is a change over time.

[0106] Further, in the ink liquid level height measuring apparatus **440**, the Hall sensor **446** includes the magnetic body **447** at the housing of the Hall sensor **446** as described above, and thus, it is possible to measure the liquid level height of the ink I with much higher accuracy.

[Correction 2]

[0107] Correction 2 (entire-region correction) which is another example of correction will be described with reference to FIGS. **6** to **7**D together with FIG. **1**.

[0108] In the above-described correction 1, the ink liquid level height measuring apparatus **440** may further include, in consideration of a case where the difference Δb between the initial value b0 and the current value bt becomes equal to or more than a predetermined value, an acquisition section that acquires a relational expression for correcting the liquid level height in the entire liquid level height region. Here, the correction section corrects the liquid level height by using the relational expression acquired by the acquisition section.

[0109] Here, as an example, in a case where the difference Δb becomes 0.5% (the predetermined value) or more with respect to the initial value b0, that is, in a case where $\Delta b/b0\times100=(b0-bt)/b0\times100\geq0.5$, the acquisition section acquires a relational expression in which the entire-region correction can be performed.

[0110] As can be seen from FIGS. **4**A and **4**B, a drift of the sensor output value of the Hall sensor **446** includes a change in the vertical axis direction and a change in the inclination. In a case where a change in the vertical axis direction, that is, the difference Δb described above becomes larger than the predetermined value, the measurement accuracy of the liquid level height of the ink I is affected due to the influence of the change in the inclination, and thus, the acquisition section acquires a relational expression in which the entire-region correction can be performed.

[0111] FIG. **6** is a flowchart illustrating entire-region correction of the Hall sensor **446** of the ink liquid level measuring apparatus **440**. FIG. **7**A illustrates initial states of the first sub-tank **411** and the second sub-tank **431** in the entire-region correction described in FIG. **6**. FIG. **7**B illustrates a state in which liquid level alignment is performed in the first sub-tank **411** and the second sub-tank **431** in the entire-region correction described in FIG. **6**. FIG. **7**C illustrates a state in which the first sub-tank **411** is hermetically sealed in the entire-region correction described in FIG. **6**. FIG. **7**D illustrates a state in which the ink I is being fed from the first sub-tank **411** to the second sub-tank **431** in the entire-region correction described in FIG. **6**.

[0112] Note that, in FIG. **6**, the first sub-tank **411** is denoted by "1ST", and the second sub-tank **431** is denoted by "2ST". Further, in FIGS. **7**A to **7**D, descriptions of some configurations of the first sub-tank section **410**, the liquid feed section **420**, and the second sub-tank section **430** are omitted, and a description of the head module **50** is also omitted. Further, the reference heights in the reference height detection sections D**1** and D**2** here are the lower limit of the first sub-tank **411** and the lower limit of the second sub-tank **431**, respectively.

[0113] As an initial state of the entire-region correction to be described below, both the liquid level heights of the ink I in the first sub-tank **411** and the second sub-tank **431** are between the upper and lower limits as illustrated in FIG. **7**A. At this time, in the first sub-tank **411**, the valve **417** (see FIG. **1**) is in an open state by the control of the measurement control section **448** (the acquisition section), and the upper space of the first sub-tank **411** is in a state of being opened to the atmosphere. In addition, in the second sub-tank **431**, the pneumatic pump **432** (see FIG. **1**) controls the pressure in the upper space of the second sub-tank **431** to a desired pressure (for example, −3 kPa) by the control of the measurement control section **448** (the acquisition section).

(Step S**21**)

[0114] The measurement control section **448** (the acquisition section) aligns each of the liquid level heights of the ink I in the first sub-tank **411** and the second sub-tank **431** with the lower limit as illustrated in FIG. **7**B.

[0115] For example, the measurement control section **448** (the acquisition section) drives the liquid feed pump **423** to feed the ink I from the first sub-tank **411** to the second sub-tank **431**, and aligns the liquid level height of the ink I in the first sub-tank **411** with the lower limit by using the reference height detector D**1** (see FIG. **1**). Thereafter, the measurement control section **448** (the acquisition section) ejects the ink I from the second sub-tank **431** by using the head module **50**, and aligns the liquid level height of the ink I in the second sub-tank **431** with the lower limit by using the reference height detection section D**2** (see FIG. **1**).

[0116] In the first sub-tank **411**, when the liquid level height of the ink I deviates from the lower limit, a relational expression L=f(P) which will be described later with reference to FIG. **8**A changes, and the entire-region correction becomes inaccurate. Accordingly, it is important to align the liquid level height of the ink I in the first sub-tank **411** with the lower limit, and any other reference height detection apparatus may also be used in addition to the reference height detection section D**1** or instead of the reference height detection section D**1**.

[0117] Further, in the second sub-tank **431**, when the liquid level height of the ink I deviates from the lower limit, the liquid level height deviates from the true value. Accordingly, it is also important to align the liquid level height of the ink I in the second sub-tank **431** with the lower limit, and any other reference height detection apparatus may also be used in addition to the reference height detection section D**2** or instead of the reference height detection section D**2**.

(Step S**22**)

[0118] The measurement control section **448** (the acquisition section) causes the valve **417** (see FIG. **1**) to be in the closed state to hermetically seal the first sub-tank **411** as illustrated in FIG. **7**C.

(Step S**23**)

[0119] After hermetically sealing the first sub-tank **411**, the measurement control section **448** (the acquisition section) drives the liquid feed pump **423** to feed the ink I from the first sub-tank **411** to the second sub-tank **431**.

(Step S**24**)

[0120] The measurement control section **448** (the acquisition section) acquires the pressure value in the hermetically sealed first sub-tank **411** by using the pressure sensor **418** (see FIG. **1**), and acquires the sensor output value of the Hall sensor **446** (see FIG. **1**).

[0121] Although details will be described later with reference to FIG. **8**A, the pressure value in the first sub-tank **411** changes in association with a change in the liquid level height of the ink I in the first sub-tank **411**, and the sensor output value of the Hall sensor **446** also changes in association with a change in the liquid level height of the ink I in the second sub-tank **431**. By using the pressure value and the sensor output value which change in this way, a relational expression to be described later is obtained and the entire-region correction is performed.

(Step S**25**)

[0122] The measurement control section **448** (the acquisition section) confirms whether the liquid level height of the ink I in the second sub-tank **431** is the upper limit. In a case where the liquid level height of the ink I in the second sub-tank **431** is the upper limit (YES), the processing proceeds to step S**26**, and in a case where the liquid level height of the ink I in the second sub-tank **431** is not the upper limit, that is, is less than the upper limit (NO), steps S**24** and S**25** are repeated.

[0123] As for the upper limit of the second sub-tank **431**, for example, a reed switch similar to the reed switch **445** used for the reference height detector D**2** may be provided in the position of the upper limit, and it may be confirmed by using the reed switch whether the liquid level height of the ink I in the second sub-tank **431** is the upper limit.

[0124] In this way, as illustrated in FIG. **7**D, the measurement control section **448** (the acquisition section) changes the liquid level height of the ink I in the second sub-tank **431** from the lower limit to the upper limit. At this time, the measurement control section **448** (the acquisition section) acquires the pressure value in the first sub-tank **411** and the sensor output value of the Hall sensor **446** of the second sub-tank **431** in association with the liquid level change from the lower limit to the upper limit.

[0125] When the ink I is fed from the first sub-tank **411** to the second sub-tank **431**, the liquid level in the first sub-tank **411** lowers. Since the first sub-tank **411** is hermetically sealed, the volume of the air increases by the volume of the fed ink I, and the pressure value in the first sub-tank **411** decreases. On the other hand, in association with the feeding of the ink I, the liquid level in the second sub-tank **431** increases, and the sensor output value of the Hall sensor **446** of the second sub-tank **431** increases. The measurement control section **448** (the acquisition section) acquires such a change in the pressure value and a change in the sensor output value in association with each other.

(Step S**26**)

[0126] The measurement control section **448** (the acquisition section) turns off the liquid feed pump **423** (also closes the liquid feed valve **424**) to stop feeding the ink I from the first sub-tank **411** to the second sub-tank **431**.

(Step S**27**)

[0127] The measurement control section **448** (the acquisition section) obtains a relationship (relational expression) between the sensor output value and the liquid level height in the second sub-tank **431** based on the pressure value in the first sub-tank **411** and the sensor output value of the Hall sensor **446** of the second sub-tank **431**.

[0128] Here, the relational expression obtained by the entire-region correction will be described with reference to FIGS. **8**A and **8**B. FIG. **8**A illustrates the relationships of the liquid level and the sensor output value in the second sub-tank **431** to the pressure value in the first sub-tank **411** in the entire-region correction illustrated in FIG. **6**. FIG. **8**B is a graph illustrating the relationship (relational expression) between the sensor output value and the liquid level, which is obtained from the relationships illustrated in FIG. **8**A. Note that, even in FIGS. **8**A and **8**B, the first sub-tank **411** is denoted by “1ST”, and the second sub-tank **431** is denoted by “2ST”.

[0129] First, by substantially the same method as the method described with reference to FIGS. **6** and **7**A to **7**D, a relational expression of the liquid level height L in the second sub-tank **431** with respect to a pressure value P in the first sub-tank **411** is obtained in advance by using an experimental value or a calculated value. That is, while the sensor output value of the Hall sensor **446** of the second sub-tank **431** is acquired in the method described with reference to FIGS. **6** and **7**A to **7**D, the liquid level height L in the second sub-tank **431** is obtained here instead by experimentation or calculation. Thus, the relational expression $L=f(P)$ illustrated in FIG. **8**A, that is, the relational expression of the liquid level height L in the second sub-tank **431** with respect to the pressure value P in the first sub-tank **411** is obtained. The measurement control section **448** stores the relational expression $L=f(P)$ in the storage section.

[0130] The measurement control section **448** performs the method described with reference to FIGS. **6** and **7**A to **7**D, that is, the entire-region correction, in a state in which the relational expression $L=f(P)$ is stored in the storage section. In FIG. **8**A, the “initial state” is the state illustrated in FIG. **7**A. Further, the “liquid level alignment” is the state illustrated in step S**21** in FIG. **6**, and FIG. **7**B. Further, the “hermetically sealing” is the state illustrated in step S**22** in FIG. **6**, and FIG. **7**C. Further, the “feeding” is the state illustrated in steps S**23** to S**26** in FIG. **6**, and FIG. **7**D. At the time of the “feeding”, the relational expression $P=g(V)$ illustrated in FIG. **8**A, that is, the relational expression of the sensor value V of the Hall sensor **446** of the second sub-tank **431** with respect to the pressure value P in the first sub-tank **411** is obtained.

[0131] Then, the measurement control section **448** (the acquisition section) obtains the relational expression $L=f(g(V))$ by using the relational expression $L=f(P)$ stored in advance in the storage section and the relational expression $P=g(V)$ obtained by the entire-region correction. That is, the measurement control section **448** (the acquisition section) obtains the relational expression $L=f(g(V))$ of the sensor-output value V of the Hall sensor **446** of the second sub-tank **431** with respect to the liquid level height L in the second sub-tank **431** (see FIG. **8**B). By this relational expression $L=f(g(V))$, it is possible to calculate a more accurate liquid level height in the range between the lower and upper limits of the second sub-tank **431**.

[0132] As described above, the measurement control section **448** (the acquisition section) obtains the relational expression $L=f(g(V))$. In addition, even when there is a change over time, the measurement control section **448** (the correction section) performs the entire-region correction in the range between the lower and upper limits of the second sub-tank **431** by using the relational expression $L=f(g(V))$, and thus, it is possible to calculate a more accurate liquid level height.

[0133] Note that the correction 2 may be executed without executing the correction 1. However, the correction 2 involves consumption of the ink I. For this reason, it is desirable, in consideration of the consumption of the ink I, to perform the correction 1, whereas in a case where the difference Δb becomes equal to or more than the predetermined value, it is desirable to perform the correction 2.

[Abnormality Determination Method]

[0134] Even when there is a change over time, an accurate liquid level height can be measured by the above-described correction 1 or 2. By using this, the ink liquid level height measuring apparatus **440** may include a determination section that performs abnormality determination of the liquid feed section **420**.

[0135] For example, while the ink liquid level height measuring apparatus **440** measures the liquid level height of the ink I in the second sub-tank **431**, the measurement control section **448** (the determination section) aligns (lowers) the liquid level height with (to) the target height by ejecting the ink I. This target height may be a height different from the above-described reference height (for example, the lower limit), and can be set to an appropriate height as long as the liquid level height in the second sub-tank **431** after the feeding of the ink I does not exceed the upper limit of the second sub-tank **431**.

[0136] Note that, in a case where the target height is the above-described reference height (for example, the lower limit), the above-described correction 1 may be executed at the time of the determination at the determination section.

[0137] After the liquid level height of the ink I in the second sub-tank **431** is aligned with the target height, the measurement control section **448** (the determination section) drives the liquid feed pump **423** for a predetermined time (for example, 10 seconds) to feed the ink I from the first sub-tank **411** to the second sub-tank **431**. As the predetermined time for which the liquid feed pump **423** is driven, an appropriate time is settable as long as the liquid level height in the second sub-tank **431** after the feeding of the ink I does not exceed the upper limit of the second sub-tank **431**.

[0138] With the above-described procedure, the measurement control section **448** (the determination section) can obtain the liquid level height with respect to the target height or the rate of change in the liquid level height for the driving time of the liquid feed pump **423**. Then, the measurement control section **448** (the determination section) can determine the state of the liquid feed section **420** (the tank supply path **421**) based on the liquid level height with respect to the reference height or the rate of change in the liquid level height.

[0139] FIG. **9** is a graph illustrating the relationship between the driving time of the liquid feed pump **423** and the liquid level height of the ink I in the second sub-tank **431** from a state in which the ink supply section **40** is a new product to a state in which the ink supply section **40** is determined to have a failure.

[0140] As illustrated in FIG. **9**, the liquid level height of the ink I in the second sub-tank **431** increases linearly with respect to the driving time of the liquid feed pump **423**, but the rate of change in the liquid level height decreases from the state in which the ink supply section **40** is a new product to the state in which the ink supply section **40** is determined to have a failure.

[0141] For example, the liquid feed section **420** in the ink supply section **40** includes the filter **425** for filtering the ink I. Since clogging progresses over time in the filter **425**, the feed amount of the ink I fed from the first sub-tank **411** to the second sub-tank **431** also decreases over time.

[0142] Accordingly, as described above, the measurement control section **448** (the determination section) drives the liquid feed pump **423** for a predetermined time to feed the ink I from the first sub-tank **411** to the second sub-tank **431**. Then, the measurement control section **448** (the determination section) determines the state of the liquid feed section **420** (whether the liquid feed section **420** has a failure) based on the liquid level height with respect to the reference height (for example, the position before the feeding) or the rate of change in the liquid level height.

[0143] FIG. **10** is a graph illustrating a determination result obtained by making determination based on a change over time in the liquid feed amount of the ink I supplied for a predetermined time. In the present embodiment, the ink liquid level height measuring apparatus **440** measures the liquid level height as a continuous value, unlike the apparatuses disclosed in PTLs 1 and 2 that detect the liquid level height discretely. For this reason, the ink liquid level height measuring apparatus **440** can determine the degree of deterioration of the liquid feed section **420** between the state of a new product and the state of a failure as illustrated in FIG. **10** by monitoring the liquid level height with respect to the reference height or the rate of change in the liquid level height over time (for example, every day).

[0144] Further, as illustrated in FIG. **10**, the ink liquid level height measuring apparatus **440** can also predict the degree of deterioration and the time of a failure by monitoring the liquid level height with respect to the reference height and the rate of change in the liquid level height over time. In this way, the degree of deterioration can be predicted, and thus, it is possible to prevent the liquid feed section **420** from having a failure to cause the inkjet printer **100** to stop, for example, by predicting the time of 80% deterioration and replacing a deteriorated component(s) at the predicted time of 80% deterioration. That is, it is possible to prevent the liquid feed section **420** from having a failure to cause the inkjet printer **100** to stop by replacing a deteriorated component(s) before a failure occurs.

[0145] Further, when the liquid feed pump **423** is driven for a predetermined time to feed the ink I from the first sub-tank **411** to the second sub-tank **431**, the location of a failure in the liquid feed section **420** can also be specified by changing the pump duty of the liquid feed pump **423**.

[0146] In the liquid feed section **420**, mainly the liquid sending pump **423** or the filter **425** becomes the portion of a failure. According to the knowledge of the inventor, the pump duty may be set to a relatively low value in a case where it is desired to perform deterioration determination of the liquid feed pump **423**, whereas the pump duty may be set to a relatively high value in a case where it is desired to perform deterioration determination of the filter **425**.

[0147] For example, it is assumed that the settable range of the pump duty of the liquid feed pump **423** is 20% to 100%. In this case, the pump duty is set to, for example, 20% in a case where deterioration determination of the liquid feed pump **423** is desired, whereas the pump duty is set to, for example, 100% in a case where deterioration determination of the filter **425** is desired.

[0148] This is because the resistance due to clogging of the filter **425** increases when the pump duty is set to a relatively high value since the resistance (pressure loss) of the filter **425** increases with the square of the flow rate of the ink I, and thus, deterioration determination of the filter **425** can be performed.

[0149] It is also because the influence of the resistance of the filter **425** becomes small and the influence of the deterioration of the liquid feed pump **423** becomes large when the pump duty is set to a relatively low value, and thus, deterioration determination of the liquid feed pump **423** can be performed. In addition, when the pump duty is a relatively low value, the flow rate of the ink I in the liquid feed pump **423** also becomes slow. For this reason, for example, in a case where a gap due to a sticking material or the like is formed in a check portion in the liquid feed pump **423**, the flow path resistance of the portion decreases, the backflow is likely to occur, and the liquid feed amount decreases, and thus, it is possible to perform deterioration determination of the liquid feed pump **423**.

[0150] As described above, the measurement control section **448** (the determination section) determines the state of the liquid feed section **420** based on the liquid level height with respect to the reference height or the rate of change in the liquid level height, and thus, it is possible to determine whether the liquid feed section **420** has a failure, and it is also possible to predict the degree of deterioration or the time of a failure.

[0151] Note that, the following supplementary note will be further disclosed with respect to the above description.

(Supplementary Note 1)

[0152] An ink liquid level height measuring apparatus that measures a liquid level height of ink in a container, the ink liquid level height measuring apparatus including:

[0153] a float that includes a magnet and moves in association with a change in the liquid level height;

[0154] a Hall element that detects, as a continuous value, a change in magnetic flux density due to the magnet that moves together with the float;

[0155] an acquirer that acquires a pressure value of air in a supply source container and an output value of the Hall element while supplying the ink from the supply source container in a hermetically sealed state to the container to change the liquid level height from a lower limit of the container to an upper limit of the container, and acquires, based on the pressure value and the output value, a relational expression for correcting the liquid level height in an entire region between the lower limit and the upper limit; and

[0156] a corrector that corrects the liquid level height by using the relational expression.

[0157] Any of the embodiment described above is only illustration of an exemplary embodiment for implementing the present invention, and the technical scope of the present invention shall not be construed limitedly thereby. That is, the present invention can be implemented in various forms without departing from the gist or the main features thereof.

[0158] For example, in the above-described embodiment, the measurement control section **448** functions as the above-described adjustment section, correction section, acquisition section, and determination section, but the control section **90** may be configured to function as the above-described adjustment section, correction section, acquisition section, and determination section.

[0159] Although embodiments of the present invention have been described and illustrated in detail, the disclosed embodiments are made for purpose of illustration and example only and not limitation. The scope of the present invention should be interpreted by terms of the appended claims.

What is claimed is:

1. An ink liquid level height measuring apparatus that measures a liquid level height of ink in a container, the ink liquid level height measuring apparatus comprising:
- a float that includes a magnet and moves in association with a change in the liquid level height;
- a Hall element that detects, as a continuous value, a change in magnetic flux density due to the magnet that moves together with the float; and
- at least one hardware processor, wherein
- the at least one hardware processor
  - adjusts the liquid level height to a reference height, and
  - acquires an output value of the Hall element and corrects the liquid level height based on the output value, the output value being an output value when the liquid level height is the reference height.

2. The ink liquid level height measuring apparatus according to claim **1**, wherein
- a housing of the Hall element includes a magnetic body.

3. The ink liquid level height measuring apparatus according to claim **1**, wherein
- the at least one hardware processor corrects the liquid level height based on an initial value of an output of the Hall element and a current value of the output of the Hall element, the initial value being a value when the liquid level height is the reference height.

4. The ink liquid level height measuring apparatus according to claim **3**, wherein
- the at least one hardware processor adds, as a correction term, a difference between the initial value and the current value to a calculation expression for obtaining the liquid level height to correct the liquid level height.

5. The ink liquid level height measuring apparatus according to claim **4**, wherein
- the at least one hardware processor
  - acquires, in a case where the difference is 0.5% or more with respect to the initial value, a relational expression for correcting the liquid level height in an entire region of the liquid level height, the entire region being a region between a lower limit of the liquid level height and an upper limit of the liquid level height, and
  - corrects the liquid level height by using the relational expression.

6. The ink liquid level height measuring apparatus according to claim **5**, wherein
- the at least one hardware processor acquires a pressure value of air in a supply source container and an output value of the Hall element while supplying the ink from the supply source container in a hermetically sealed state to the container to change the liquid level height in the container from the lower limit to the upper limit, and acquires the relational expression based on the pressure value and the output value.

7. The ink liquid level height measuring apparatus according to claim **1**, wherein
- the at least one hardware processor performs determination of a state of an ink supply path to the container based on the liquid level height with respect to the reference height or a rate of change in the liquid level height.

8. The ink liquid level height measuring apparatus according to claim **7**, wherein
- the at least one hardware processor, during the determination,

adjusts the liquid level height to the reference height, and

acquires an output value of the Hall element when the liquid level height is the reference height.

9. The ink liquid level height measuring apparatus according to claim **7**, wherein

the at least one hardware processor performs the determination of the state of the ink supply path based on the rate of change in the liquid level height in a case where the ink has been supplied to the container for a predetermined time.

10. An image forming apparatus, comprising:

a container that stores ink;

an image former that forms an image by using the ink supplied from the container; and

the ink liquid level height measuring apparatus according to claim **1**.

11. The image forming apparatus according to claim **10**, wherein

the at least one hardware processor, during an image formation by the image former,

adjusts the liquid level height to the reference height, and

acquires an output value of the Hall element when the liquid level height is the reference height.

12. The image forming apparatus according to claim **11**, wherein

the at least one hardware processor

supplies the ink to the container, when the liquid level height of the ink in the container becomes lower than the reference height due to the image formation by the image former, and

stops supplying the ink, when the liquid level height reaches the reference height.

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